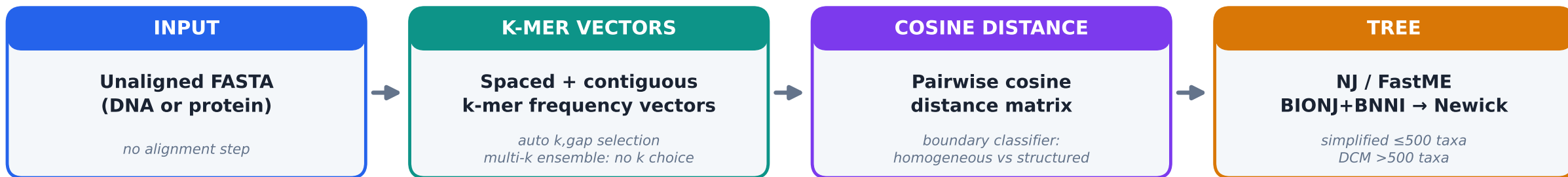


Fusang: Tardigrade Edition

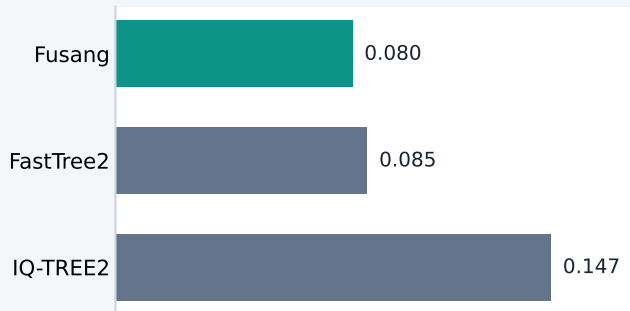
Alignment-free phylogenetic inference with k-mer frequency vector cosine distances



Validated across simulation and real data

Indel-rich accuracy

n=200, indel rate 0.02, vs true tree (nRF ↓)



matches FastTree2 (p=0.052, n.s.-borderline)

1.8× more accurate than IQ-TREE2 GTR (p<0.001)

Real-data validation

AFproject community benchmark, ETE3 nRF ↓

Fish mitogenomes (n=25, ~17 kb) **0.045**

community-benchmark ceiling — ties MSA+ML

13 mtDNA genes (168-1,866 bp) **0.451**

Fusang-FastTree2 agreement (13 loci)

SwissTree proteins (11 families) **0.239**

k-mer cosine vs Co-phylog 0.361 (p=0.006)

no k selection at any scale: 500 bp genes → 17 kb genomes

Fast, scalable, ready to use

10,000 taxa **54.4 s**

single CPU core

1,000 taxa **5.2 s**

simplified pipeline

boundary classifier **88/88**

E2E scenario routing

pre-compiled Windows/Linux binaries · MIT license